

Minghan Gao

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EDUCATION

Cornell University

Aug 2022 – May 2026

B.S., Information Science, and Communication

- GPA: 3.956
- **Minor:** Artificial Intelligence and Computer Science

RESEARCH INTERETS

1. Wearable sensing and signal processing
2. Ubiquitous and embodied computing
3. Accessibility and assistive technologies

PREPRINT & PUBLICATIONS

1. Kye Shimizu, **Minghan Gao**, Ananya Ganesh, Pattie Maes. 2025. **VoiceMorph: How AI Voice Morphing Reveals the Boundaries of Auditory Self-Recognition.** <https://doi.org/10.48550/arXiv.2510.16192> 
2. Hyunchul Lim, **Minghan Gao**, Franklin Mingzhe Li, Nam Anh Dang, Ianip Sit, Michelle M Olson, and Cheng Zhang. 2025. **Exploring the Impact of Emotional Voice Integration in Sign-to-Speech Translators for Deaf-to-Hearing Communication.** In *Proceedings of the 28th ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW'25)*. <https://doi.org/10.1145/3757597> 

RESEARCH EXPERIENCE

Egocentric VLMs for Surgical Performance Assessment and Corrective Feedback in Central Venous Catheterization

Hybrid

Sep 2026 – Now

Visiting Researcher, Professor Justin Chan, Carnegie Mellon University

- Developing a vision-language-model-based framework for egocentric surgical video understanding to assess procedural performance from head-mounted camera footage under real-world challenges such as motion blur, operator variability, and changing tool distance.
- Investigating zero-/few-shot recognition of procedural steps, active tools, and arbitrary execution errors, with the goal of comparing expert and non-expert performance beyond simple step-boundary detection.
- Designing methods for evidence-grounded error reasoning and corrective feedback, enabling the system to identify what went wrong, explain the likely cause, and provide actionable guidance during or after a procedure.

Expressive Voice Adaptation in AAC Conversation

New York, NY

Research Assistant, Professor Thijs Roumen and Tobias Weinberg (PhD), Cornell Tech

Sep 2026 – Now

- Investigating how AAC users decide when and how to adapt the expressive style of their synthetic voice during semi-structured conversations with familiar communication partners.
- Designing interaction techniques for low-cost voice adaptation, exploring alternatives to explicit tagging such as inferred suggestions, selectable options, and lightweight controls that minimize interruption to conversation.
- Studying how voice adaptations influence self-expression, partner interpretation and response, conversational flow, and interaction cost, with attention to when control burden outweighs expressive benefit.

VoiceMorph: Boundaries of Auditory Self-Recognition

Cambridge, MA

Research Intern, Professor Pattie Maes and Kye Shimizu (Master), MIT Media Lab

May 2025 – Sep 2025

- Developed a real-time experimental system in TouchDesigner integrating a deep learning voice morphing model, enabling automated accuracy and reaction-time capture to quantify perceptual boundaries of voice identity.
- Performed quantitative analysis of 2,000+ trials (21 participants), modeling perceptual thresholds and analyzing demographic, cognitive, and acoustic influences in self-voice recognition.

- Led acoustic quality control and synthesized insights from quantitative and qualitative data, ensuring valid stimuli (UTMOSv2 + demographic matching) and revealing prosodic strategies participants use to judge self-voice identity.

Emotional Voice Integration in American Sign Language (ASL) Translators

Ithaca, NY

Research Assistant, Professor Cheng Zhang and Hyunchul Lim (PhD), Cornell University

Feb 2024 – Sep 2024

- Engineered an AI-based emotional voice generation pipeline using neural TTS, synthesizing and evaluating speech across five emotional categories for ASL translation.
- Designed and deployed Qualtrics studies (N = 93 hearing, N = 48 DHH) with randomized conditions and synchronized ASL + voice stimuli to evaluate emotion comprehension.
- Processed ≈ 100 ASL videos, integrating voice overlays, timing alignment, and expression-order control.
- Led statistical and visualization analysis (Kruskal–Wallis, ANOVA, post-hoc, confusion matrices) to quantify effects of emotional voice on recognition accuracy and cognitive effort.

arXiv Reading Behavior Analysis

Ithaca, NY

Research Intern, Professor Yian Yin, Cornell University

June 2024 – Aug 2024

- Conducted large-scale data analysis of 4.8 million arXiv users to explore academic reading behaviors and referral influences, filtering out bots to focus on genuine, consistent readers.
- Designed and implemented structured user timelines integrating paper access dates, referral sources, and engagement depth, then analyzed referral-driven patterns to show how discovery pathways correlate with reading recency, team composition, and paper visibility.

Persuasive Chatbot

Ithaca, NY

Research Assistant, Professor Susan Fussell, Cornell University

Sep 2023 – Dec 2023

- Designed and implemented the web-based frontend in React (HTML/CSS), enhancing UI responsiveness and optimizing layout across devices.
- Integrated GPT-4 into the system backend via OpenAI API, enabling real-time chatbot interaction and automatic conversation logging to Google Docs for researcher review.

PROJECT

HopIn: Accessible Ride-Sharing Platform for Mobility Equity in Ithaca

Aug 2025 – Dec 2025

- Led user research with low-income and disabled riders, TCAT staff, and local advocates to identify key accessibility, safety, and equity needs in Ithaca's transit system.
- Synthesized findings into concrete design requirements and Figma prototypes for ride-matching, medical trips, and accessibility-support workflows.
- Ran expert reviews and user tests, iteratively refining interaction flows and accessibility features based on qualitative feedback.

Low-voice Whisper Recognition on Smart Watch

Jan 2025 – May 2025

- Designed and prototyped a smartwatch-based whisper speech interface using active ultrasonic sensing.
- Collected a dataset of 6,380 labeled whisper utterances from four participants across 29 command classes, using a custom 3D-printed wrist-worn device with dual ultrasonic speakers and microphones powered by a Teensy 4.1.
- Developed a deep learning pipeline using a modified four-channel ResNet-18 with data augmentation strategies (transfer learning, random cropping, time-frequency masking, and Gaussian noise augmentation), achieving 55% accuracy (macro F1: 0.56).

TEACHING EXPERIENCE

Cornell Ann S. Bowers College of Computing and Information Science

Ithaca, NY

Teaching Assistant

Aug 2023 – Dec 2023

- Guided students in CS 1110 (Introduction to Computing), instructed by Prof. Walker White during office hours, helping them navigate course concepts, assignments, and programming techniques.
- Graded student assignments, projects, and exams, offering feedback to enhance their understanding and proficiency in programming concepts.

EXTRACURRICULAR ACTIVITIES

Women in Computing at Cornell , Committee Member	<i>Aug 2024 – Dec 2024</i>
Cornell Daily Sun , Writer & Web Developer	<i>Aug 2022 – Dec 2023</i>
Cornell Student Assembly , Committee Member	<i>Sep 2023 – Dec 2023</i>
Cornell Undergraduate Research Magazine , Writer	<i>Jan 2023 – May 2023</i>

AWARDS & SCHOLARSHIPS

1. Dean's List
2. Cornell Bowers CIS Undergraduate Research Experience (BURE) Recipient (\$7,000), 2024

SKILLS

Languages: Python, Java, C++, HTML, CSS, JavaScript, SQL

Libraries & Frameworks: pandas, NumPy, scikit-learn, PyTorch, Matplotlib, seaborn, D3.js

Software: TouchDesigner, Final Cut Pro, Figma, Fusion 360, Lightroom, Inkscape, JMP

Hardware: Arduino, PulseView, Processing

Qualitative Skills: User interviews, ethnographic methods (contextual inquiry, participant observation, diary studies), thematic and narrative analysis